

ESc 301.03

The Environmental Dimension

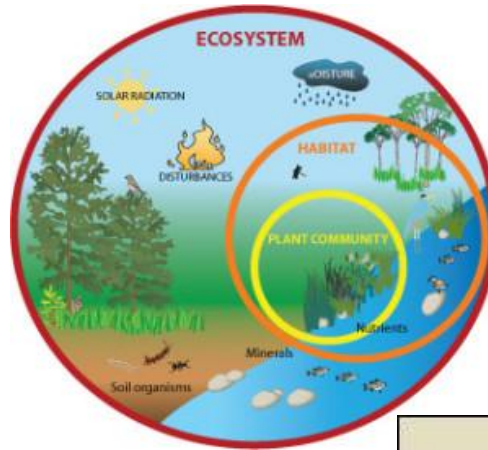
Ecosystems



Assoc. Prof. Dr. Başak Güven

Ecosystem

An ecosystem is a dynamic complex of plant, animal and microorganism communities and the non-living environment, interacting as a functional unit. Humans are an integral part of ecosystems.

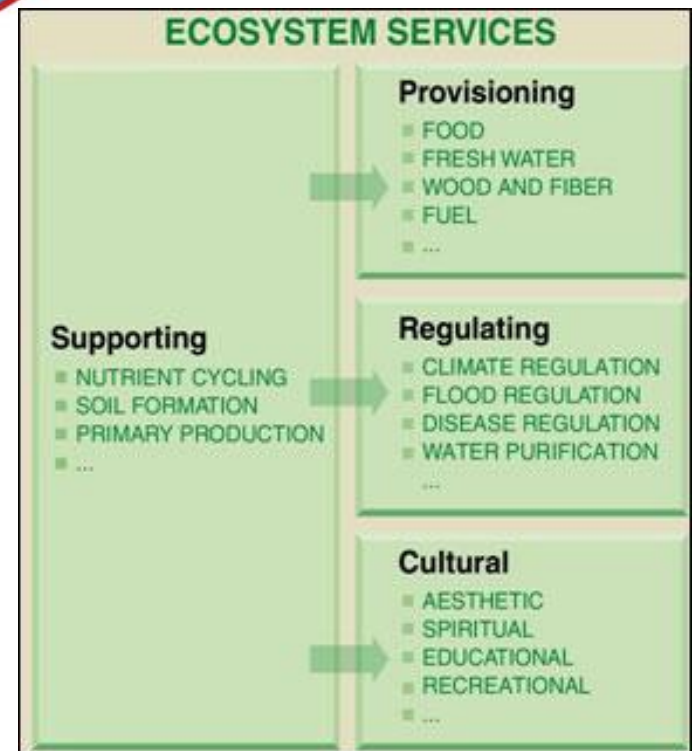


Ecosystem Goods

- ❖ Food
- ❖ Construction material
- ❖ Medicinal plants
- ❖ Tourism

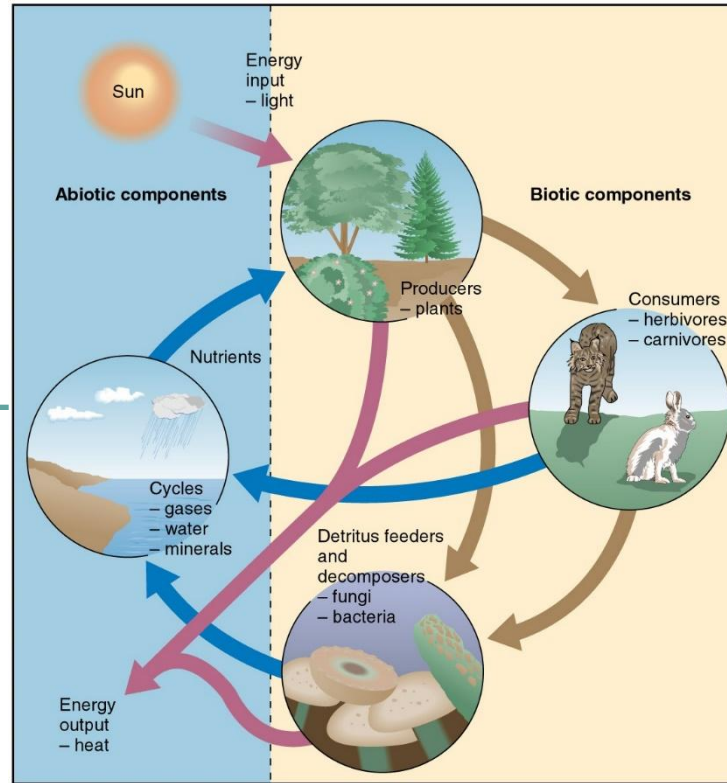
Ecosystem Services

- ❖ Maintaining hydrological cycles
- ❖ Regulating climate
- ❖ Cleansing water and air
- ❖ Maintaining the gaseous composition of the atmosphere
- ❖ Pollinating crops and other important plants
- ❖ Generating and maintaining soils
- ❖ Storing and cycling nutrients
- ❖ Absorbing and detoxifying pollutants
- ❖ Providing beauty, inspiration and recreation



Ecosystem Structure

What are the major components of an ecosystem?



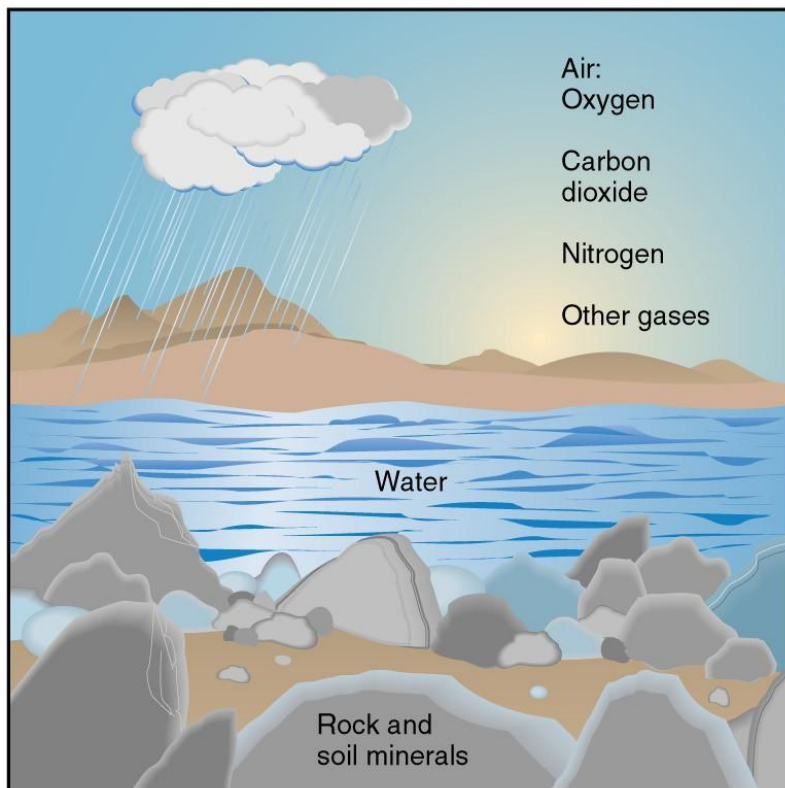
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What happens to matter and energy in ecosystems?

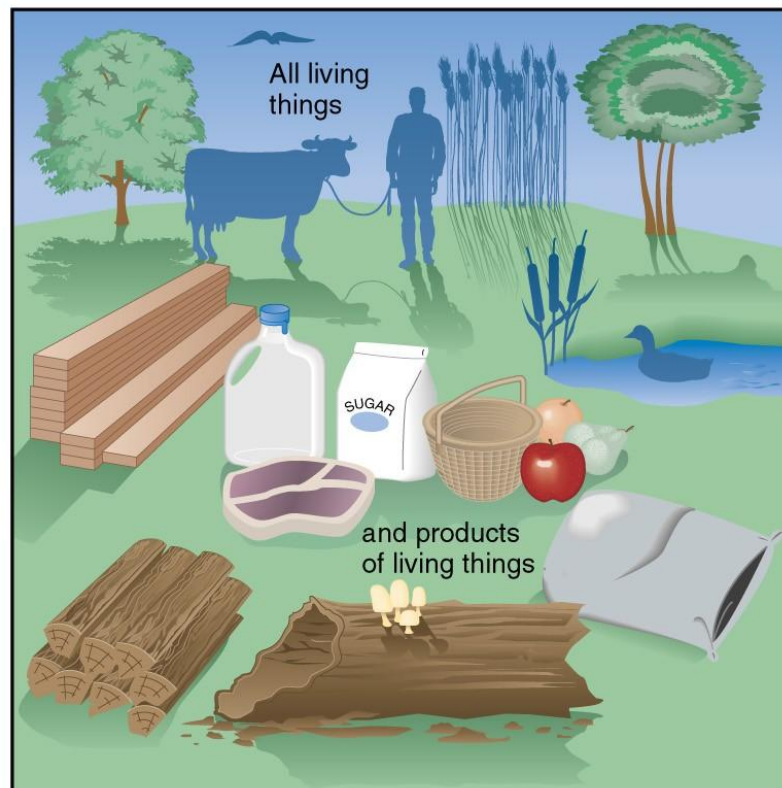
- ❖ Water, air, temperature, soil, light levels, precipitation, salinity
- ❖ Sets tolerance limits for populations and communities
- ❖ Some are limiting factors that structure the abundance of populations

- ❖ Producers, consumers, decomposers
- ❖ Plants, animals, bacteria/fungi
- ❖ Biotic interactions with biotic components include predation, competition, symbiosis, parasitism, commensalism etc.

Inorganic



Organic



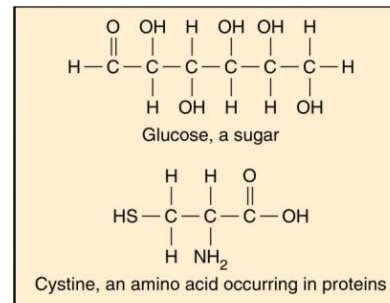
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Inorganic: molecules without carbon-carbon or carbon-hydrogen bonds

Examples: NaCl , NH_4 , H_2SO_4

Organic: carbon-based molecules

Examples: $\text{C}_6\text{H}_{12}\text{O}_6$, CH_4



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Energy & Matter in Ecosystems

Energy

anything that has the ability to move matter; has no mass and does not occupy space

- ❖ Cannot be created or destroyed
- ❖ Can be changed from one form to another
- ❖ Cannot be recycled
- ❖ Can be measured

Matter

anything that occupies space and has mass

- ❖ Cannot be created or destroyed
- ❖ Can be changed from one form into another
- ❖ Can be recycled
- ❖ Can be measured where gravity is present

Laws of Thermodynamics

First Law: Energy is neither created nor destroyed but may be converted from one form to another.

Second Law: With each successive energy transfer, less energy is available to perform work. Some energy is lost as heat.

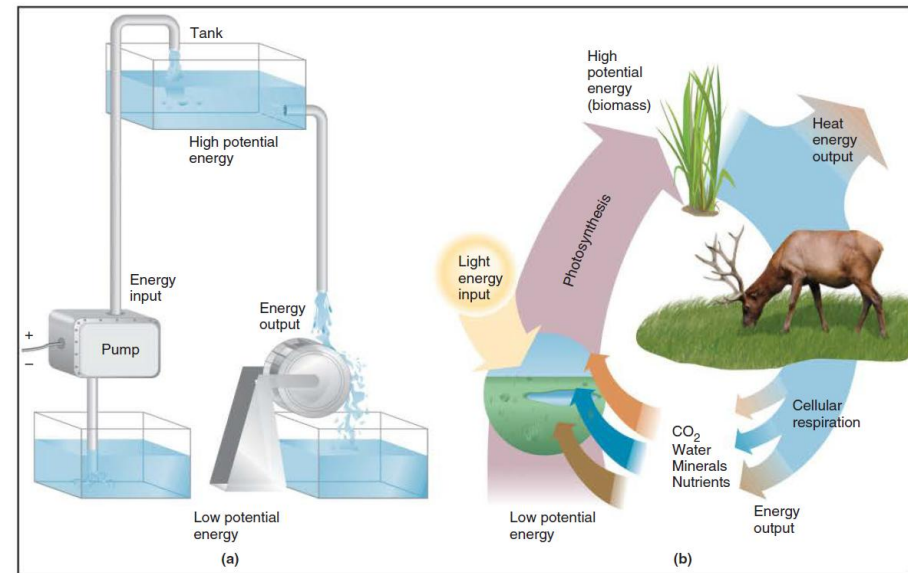


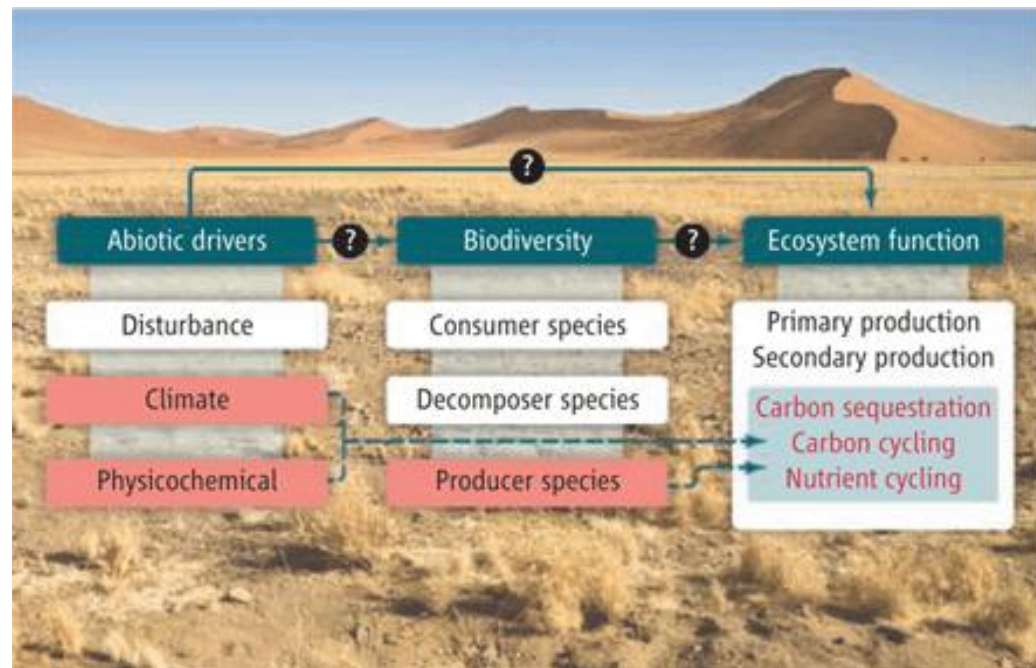
Figure 3-15 Storage and release of potential energy. (a) A simple physical example of the storage and release of potential energy. (b) The same principle applied to ecosystems.

Ecosystem Function

- ❖ Energy flows through ecosystems
- ❖ Energy must be supplied from an external source
- ❖ Energy flows in a one-way path through living systems and is used for work or lost as heat
- ❖ Chemical matter (nutrients) cycles within or/or between ecosystems

Biogeochemical Cycles

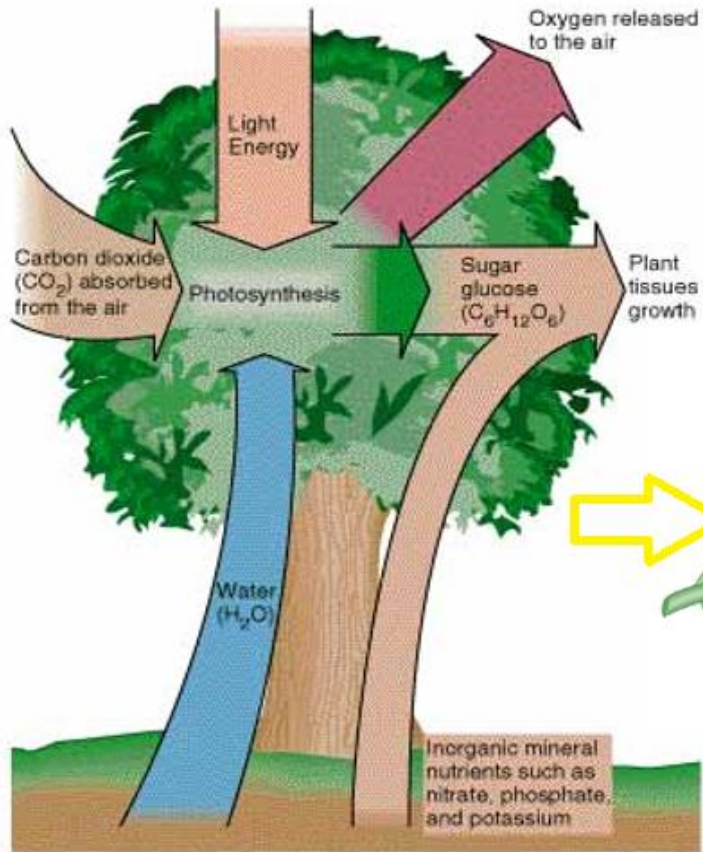
- Carbon
- Nitrogen
- Phosphorus



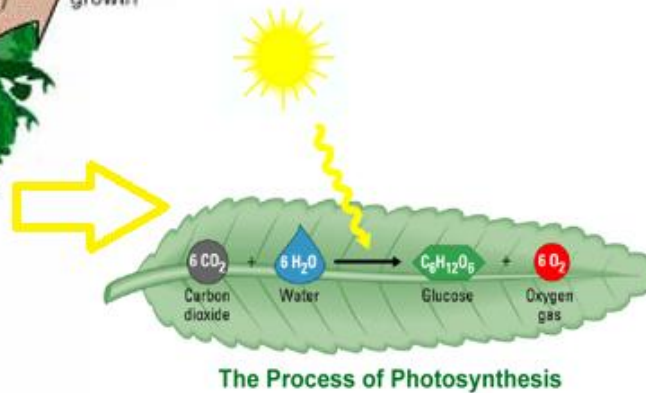
Energy and Matter in Ecosystems

- ❖ Most of the important relationships between living organisms and the environment are controlled directly or indirectly by the amount of available incoming energy received at the Earth's surface from the sun.
- ❖ A tiny fraction of the solar energy is captured by photosynthesizing organisms. Most ecosystems fix less than 1 % of the sunlight available for photosynthesis.
- ❖ Living organisms can use energy in basically two forms: radiant or fixed.
- ❖ Radiant energy exists in the form of electromagnetic energy, such as light. Radiant energy is used by plants in the process of photosynthesis.
- ❖ Fixed energy is the potential chemical energy found in organic substances. Once fixed by plants, organic energy can move within ecosystems. Fixed energy is used by all organisms in the process of respiration.

Ecosystems use sunlight as their source of energy



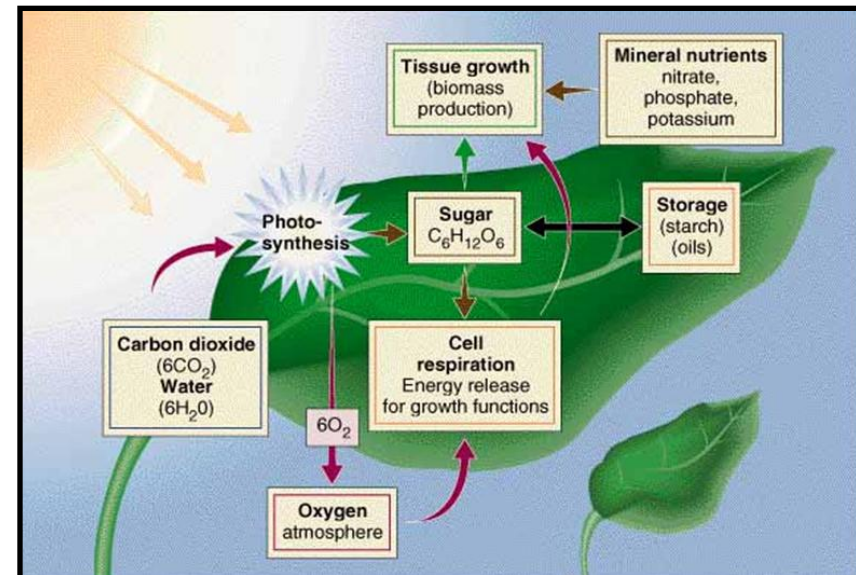
Photosynthesis is the process of converting solar energy, water, and carbon dioxide into carbohydrates and oxygen. The process occurs in two steps: first light energy is absorbed by chlorophyll to split a molecule of water releasing hydrogen and oxygen. The second step uses the energy to convert carbon dioxide to carbohydrates.



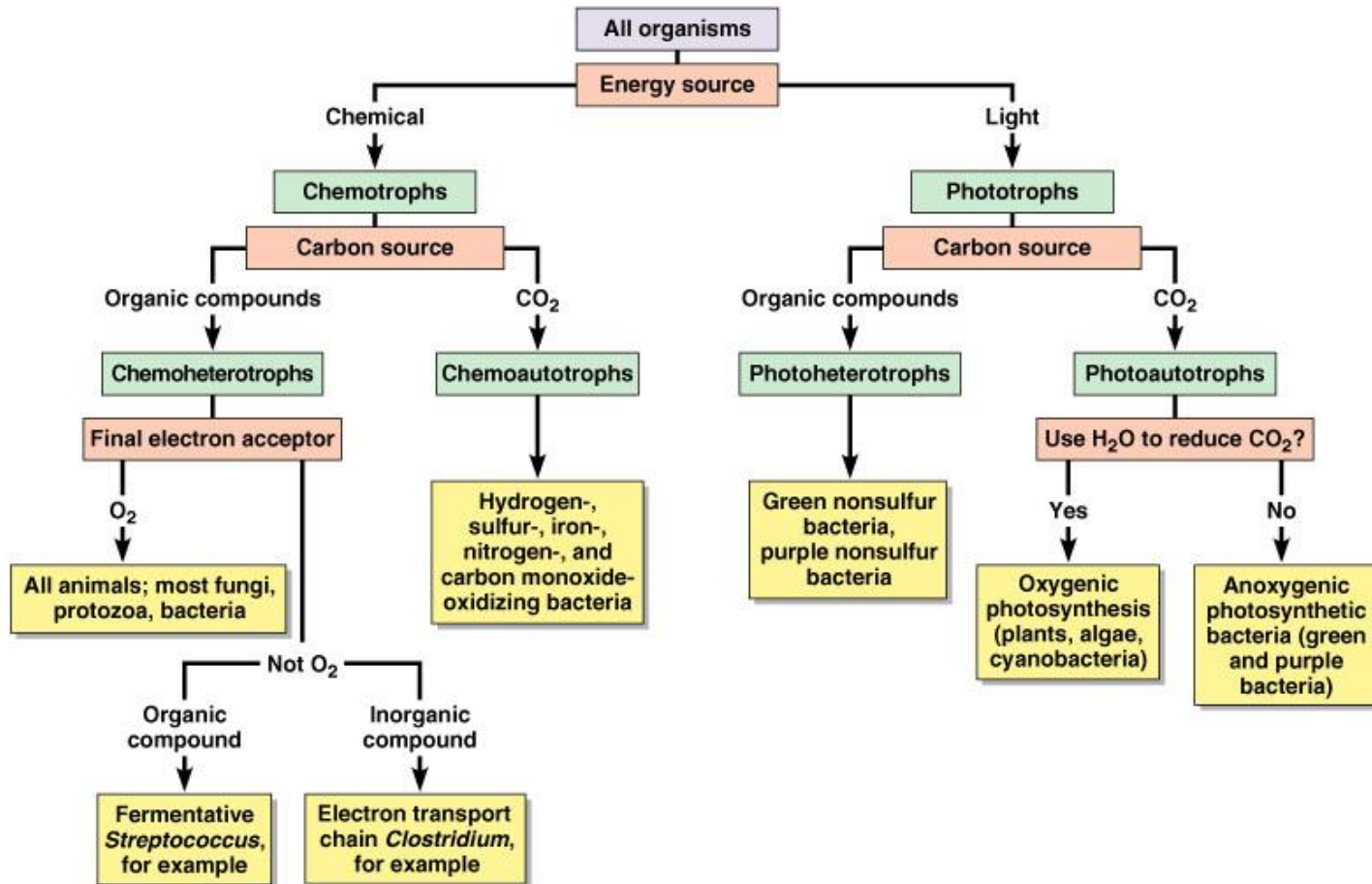
Photosynthesis



Respiration



Energy and Matter Flow in Ecosystems

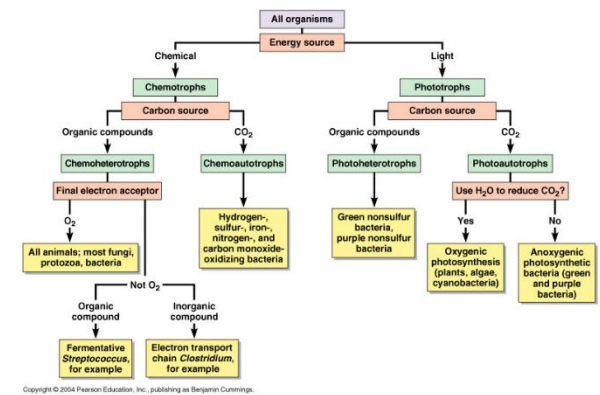


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Heterotrophs: Organisms that require fixed energy found in organic molecules for their survival (the animals).

Autotrophs (self-feeders): Organisms that can take energy from inorganic sources and fix it into energy rich organic molecules. If this energy comes from light then these organisms are called photosynthetic autotrophs (the plants).

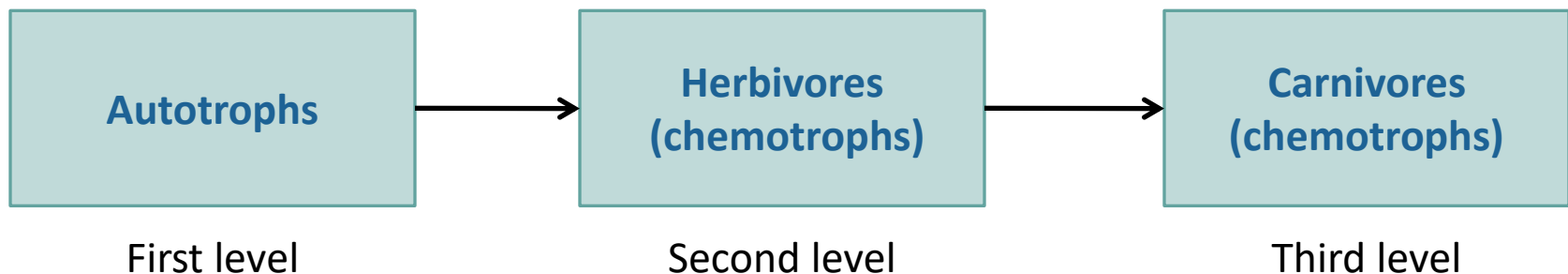
Some Important Definitions



- ❖ Sunlight using organisms are called primary producers.
- ❖ Primary producers also obtain their carbon from inorganic sources such as carbon dioxide and bicarbonate are called autotrophic.
- ❖ The rate of production of biomass glucose, cells, and other organic chemicals by the primary producers is referred to as net primary productivity.
- ❖ Aerobic respiration is simply the breakdown of organic chemicals, such as sugars and starches, by molecular oxygen to form gaseous carbon dioxide.
- ❖ Organisms that are able to obtain energy through photosynthesis but are not capable of reducing carbon dioxide and obtaining carbon from reduced carbon compounds generated by other organisms are called photoheterotrophs.
- ❖ The chemotrophs obtain their energy from organic or inorganic carbon rather than from light.

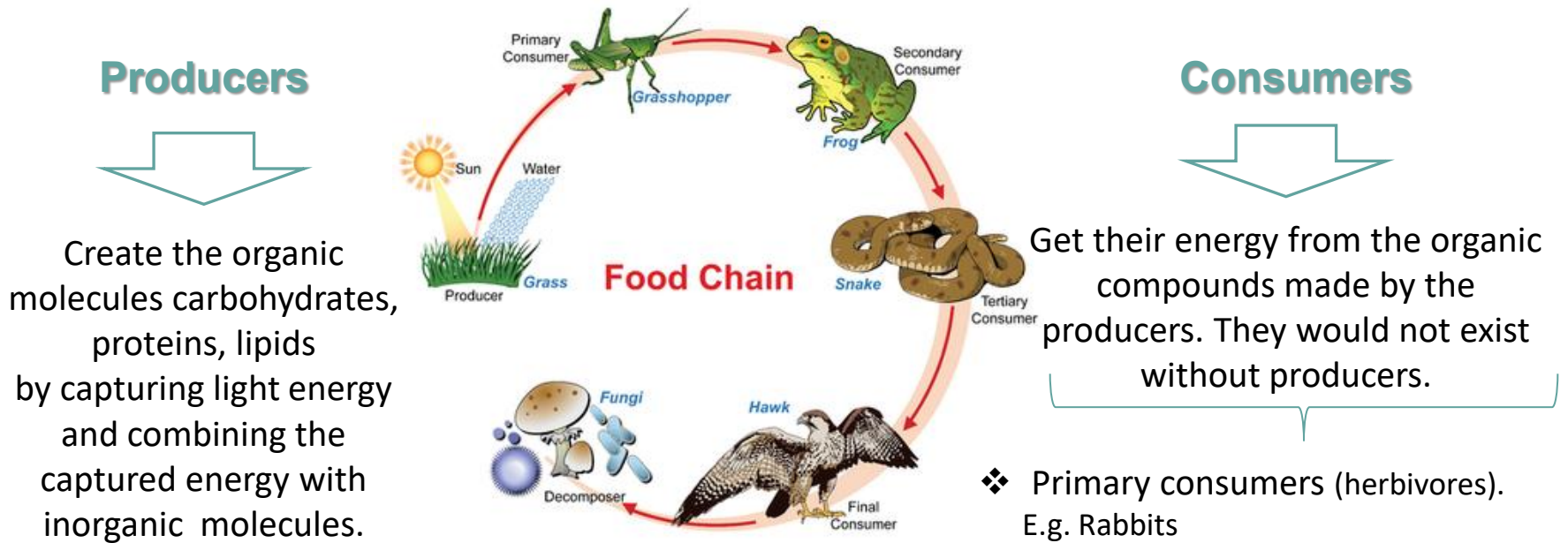
- ❖ Heterotrophs who obtain their energy from living organisms are called consumers.
- ❖ Consumers that consume plants are known as herbivores.
- ❖ Consumers that consume other animals are known as carnivores.

As we move up the trophic level from the primary producers, we find those organisms that are known as the **primary consumers**. These **chemoheterotrophic** organisms are the herbivores that eat plant material. Although chemoheterotrophs may obtain energy from chemicals formed by other organisms, under most circumstances they do not consume the organism to obtain those compounds. Rather the compounds they consume were excreted by the living organism or released during the decay of the dead organisms. These organisms are often referred to as the **decomposers**. The **secondary consumers**, also chemoheterotrophic organisms, are carnivores that eat the flesh of animals.



What we have just described is the **food chain**, a description of the complex relationship between organisms in an ecosystem.

Organisms can be either producers or consumers in terms of energy flow through an ecosystem.



Detritus feeders (detritivores) and decomposers

are heterotrophs that obtain their energy either from dead organisms or their remains.



- ❖ Detritivores are invertebrates that feed on organic wastes and dead organisms (detritus) from all trophic levels.
- ❖ Decomposers are bacteria and fungi that break down dead material into inorganic materials.

Autotrophs

Make their own organic matter from inorganic nutrients and an environmental energy source

Heterotrophs

Must feed on organic matter for energy

Producers

Photosynthetic green plants:
use chlorophyll to absorb light energy

Photosynthetic bacteria:
use purple pigment to absorb light energy

Chemosynthetic bacteria:
use high-energy inorganic chemicals such as hydrogen sulfide

Consumers

Primary consumers/herbivores:
animals that feed exclusively on plants

Omnivores:
animals that feed on both plants and animals

Secondary consumers/carnivores:
animals that feed on primary consumers

Higher orders of consumers/carnivores:
animals that feed on other carnivores

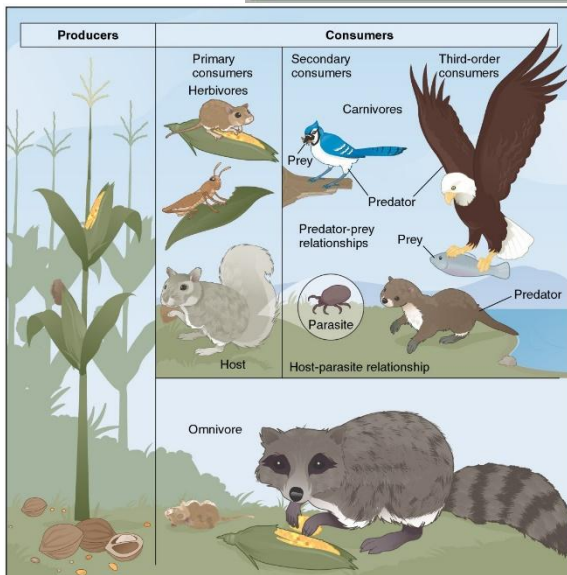
Parasites:
plants or animals that become associated with another plant or animal and feed on it over an extended period of time

Detritus feeders and decomposers:
organisms that feed on dead organic material

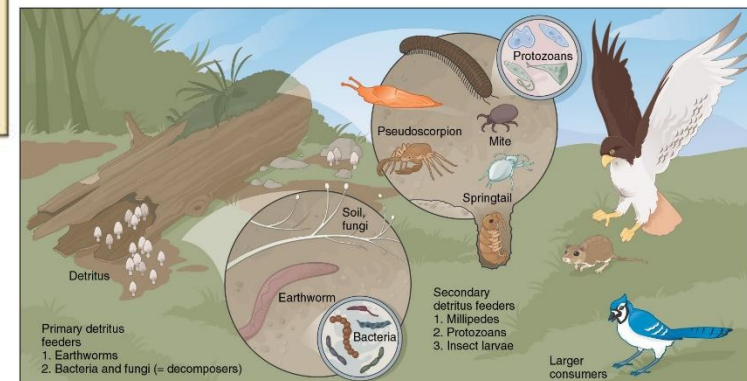
Decomposers:
fungi and bacteria that cause rotting

Primary detritus feeders:
organisms that feed directly on detritus

Secondary and higher orders of detritus feeders:
feed on primary detritus feeders

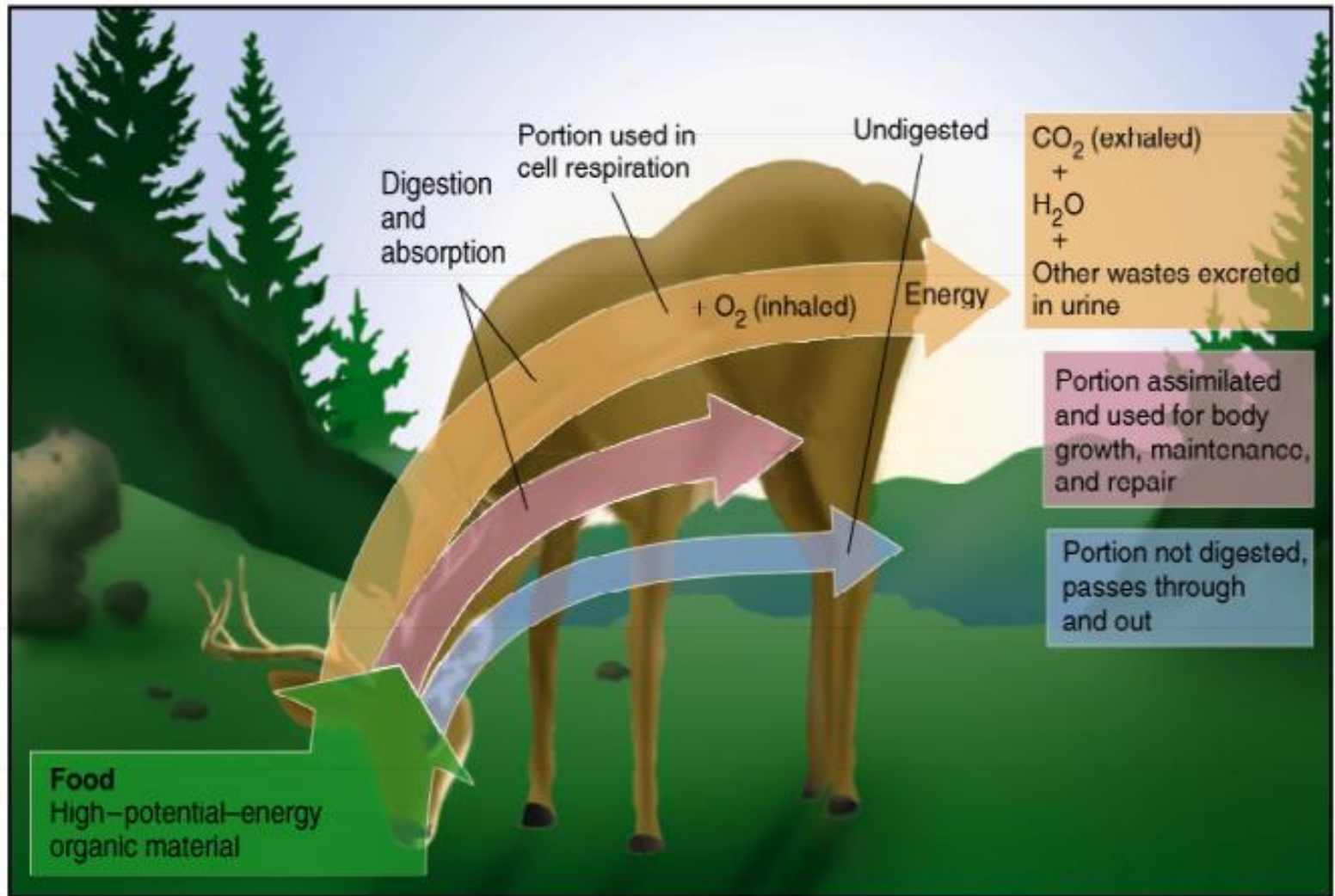


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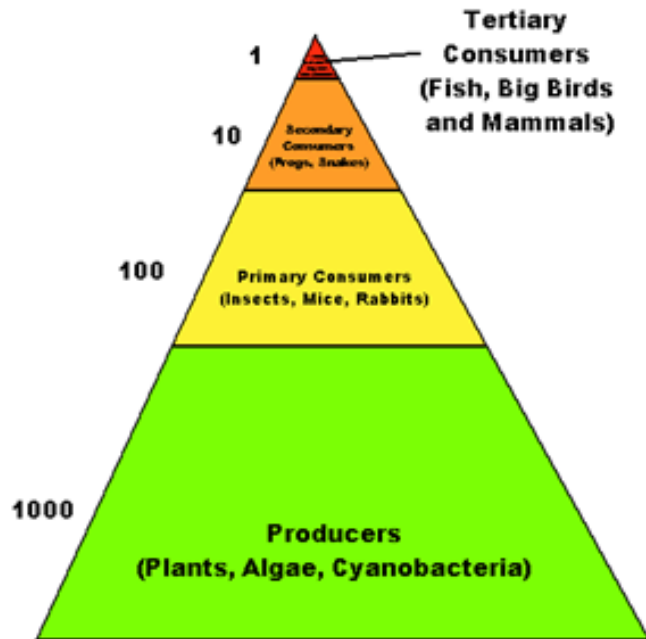
Food Assimilation in Consumers



Trophic Relationships

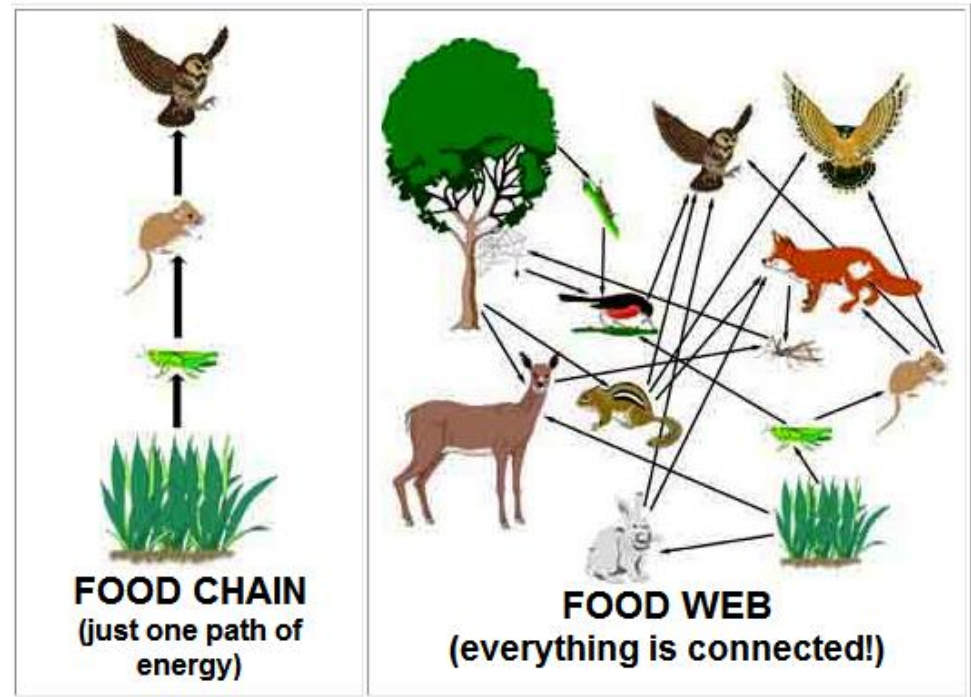
Food Chain

(A description of who eats whom)



Food Web

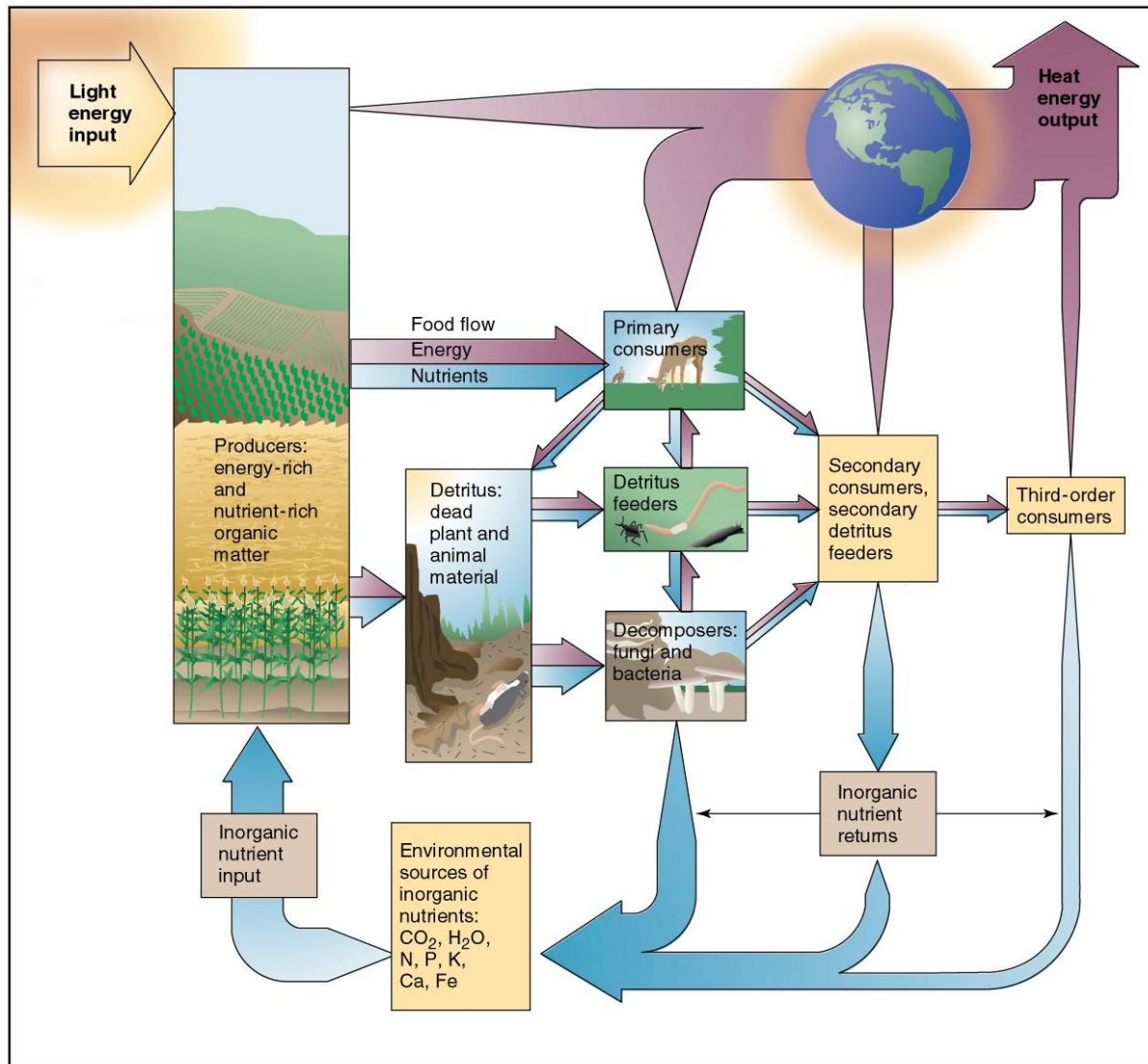
(Complexes of feeding relationships)



Trophic Levels or Feeding Levels (the position of organisms in the food chain).

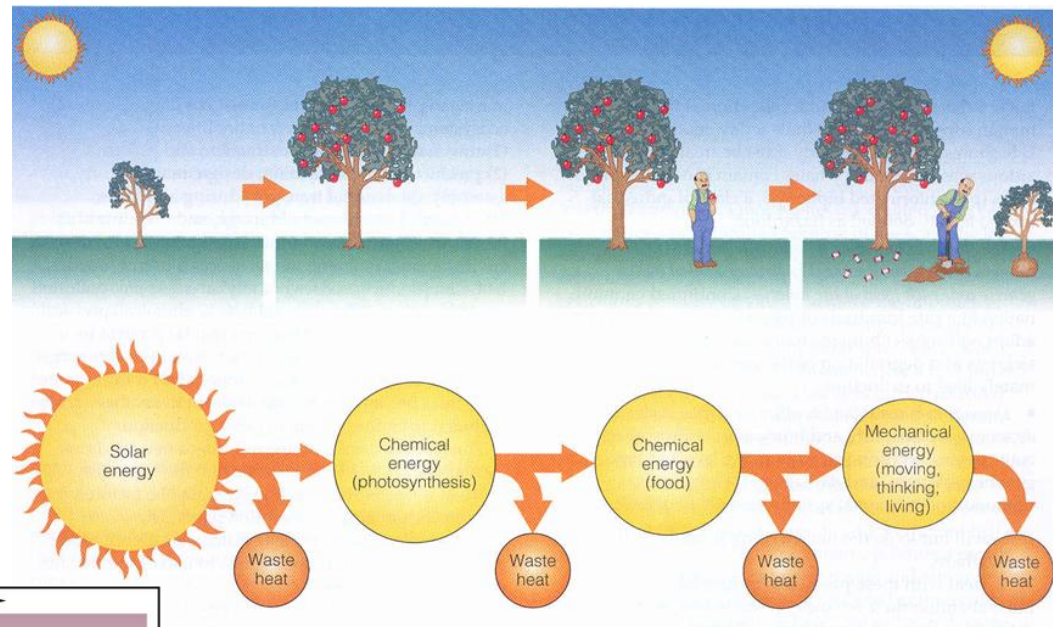
- ❖ All producers belong to the first trophic level.
- ❖ All herbivores (primary consumers) are on the second trophic level.
- ❖ All primary carnivores (secondary consumers) are on the third trophic level.

Nutrient Recycling and Energy Flow through an Ecosystem

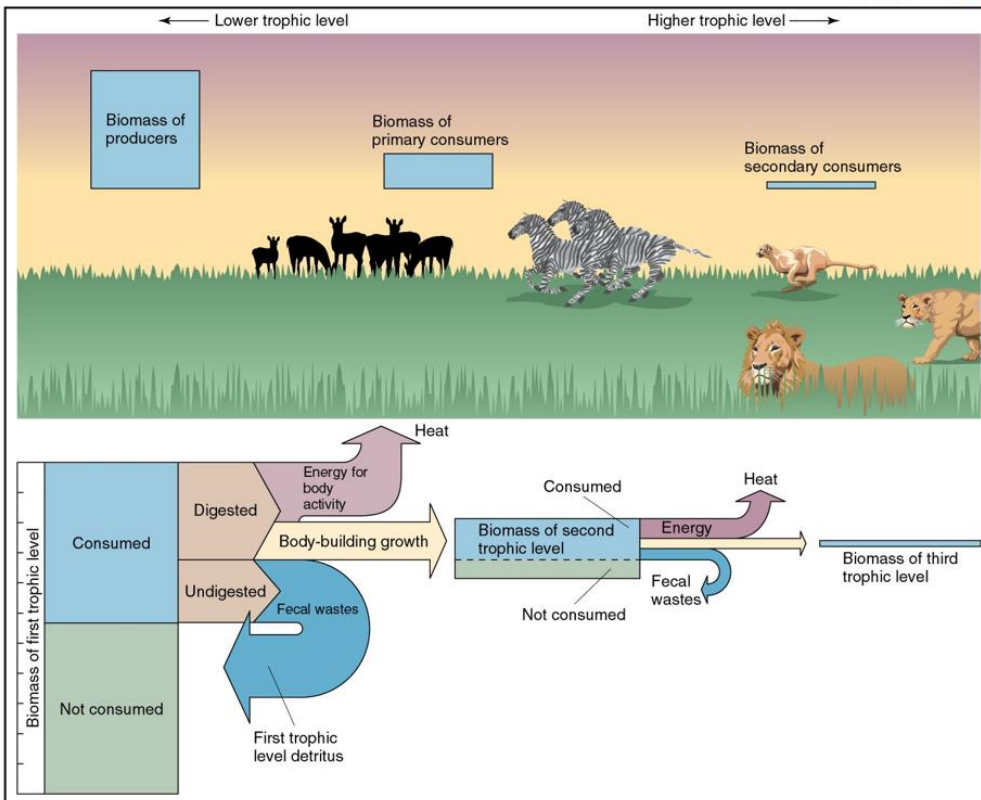


The Law of 10 %

Each time energy is changed from one form to another, some of the initial input of high quality energy is degraded, usually to low quality heat that disperses into the environment.



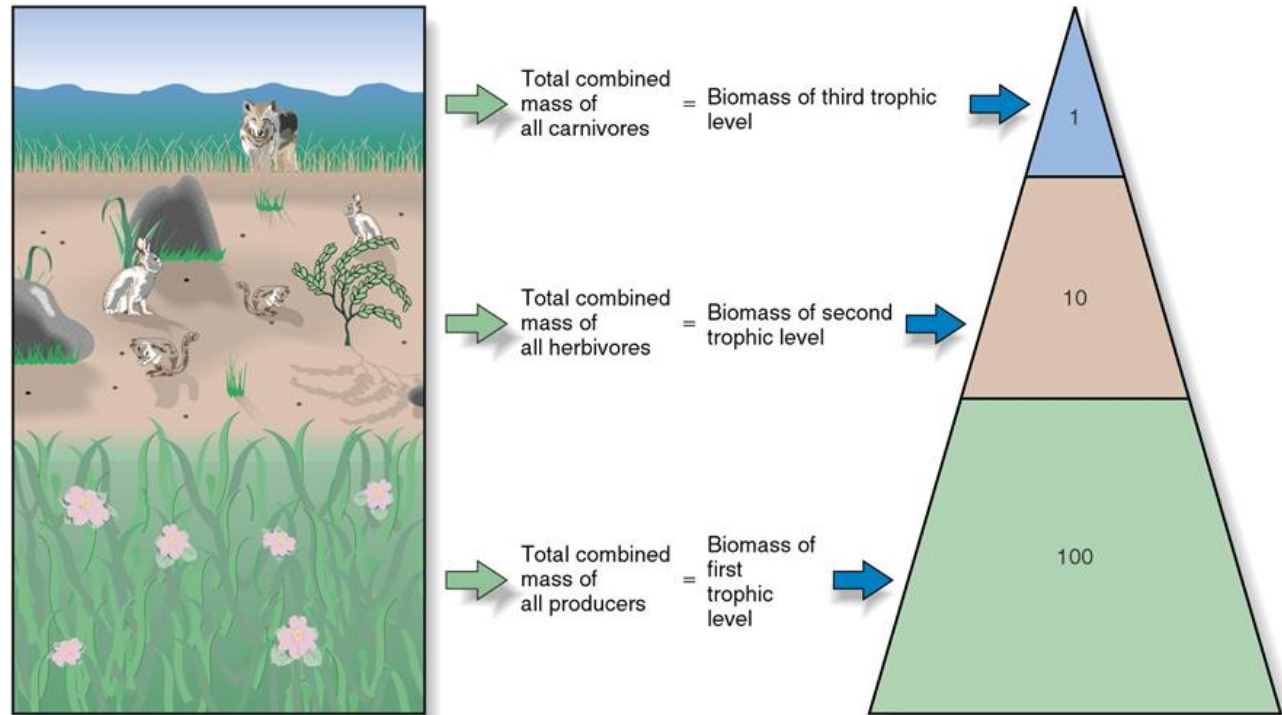
On the average about 10 % of the energy entering a feeding level (trophic level) is transferred to the next one.



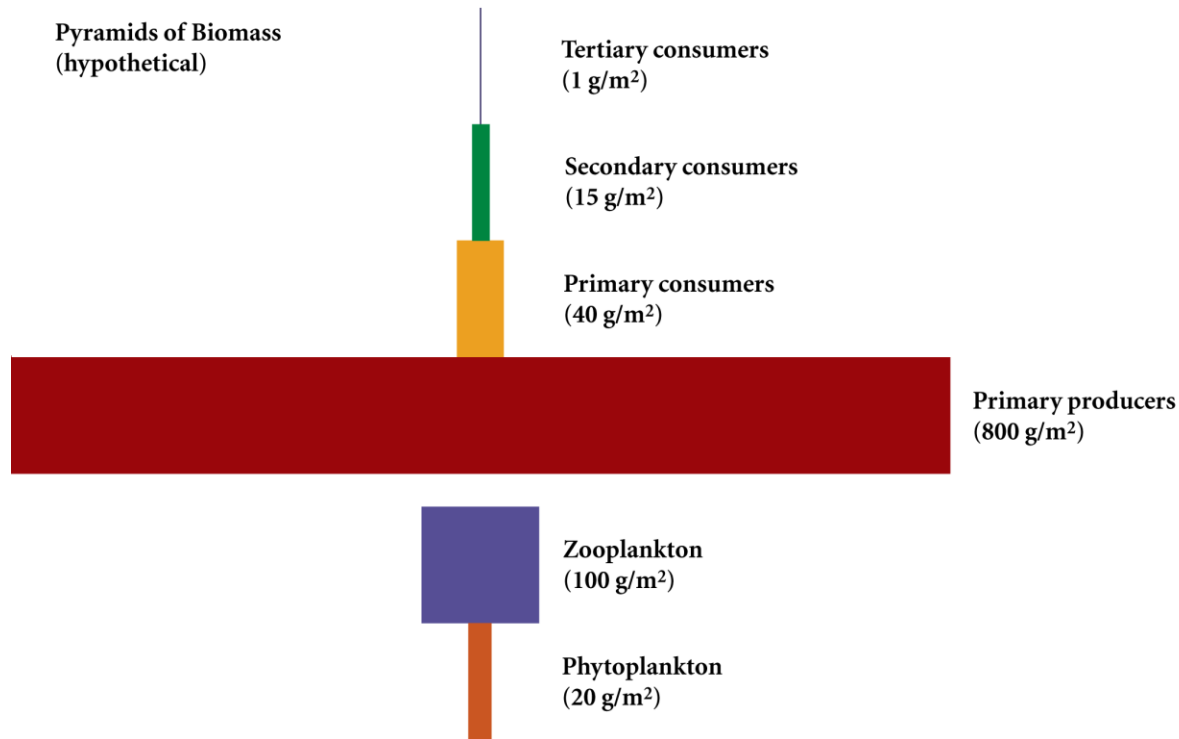
Biomass and Biomass Pyramid

The dry mass of organic material in the organism(s) is defined as **biomass**. All biomass can be arranged into a feeding relationship with the producers on the first trophic level. On average, 10% of the energy from one trophic level moves to the next trophic level. At each trophic level most of the organisms are not consumed, portions of organisms consumed pass through the consumer undigested. At each step energy is lost in the form of heat. Because so little energy can be transferred between trophic levels, it is necessary that the first trophic level contains the greatest number of organisms, and the subsequent trophic levels contain fewer and fewer organisms. Limitations on the transfer of energy between trophic levels creates the biomass pyramid.

Most ecosystems have huge number of primary producers supporting a smaller number of herbivores, supporting a smaller number of secondary consumers. If organisms (humans) eat high on the biomass pyramid (trophic levels 3, 4, 5, etc.), then fewer organisms can be supported than if organisms eat lower on the biomass pyramid.

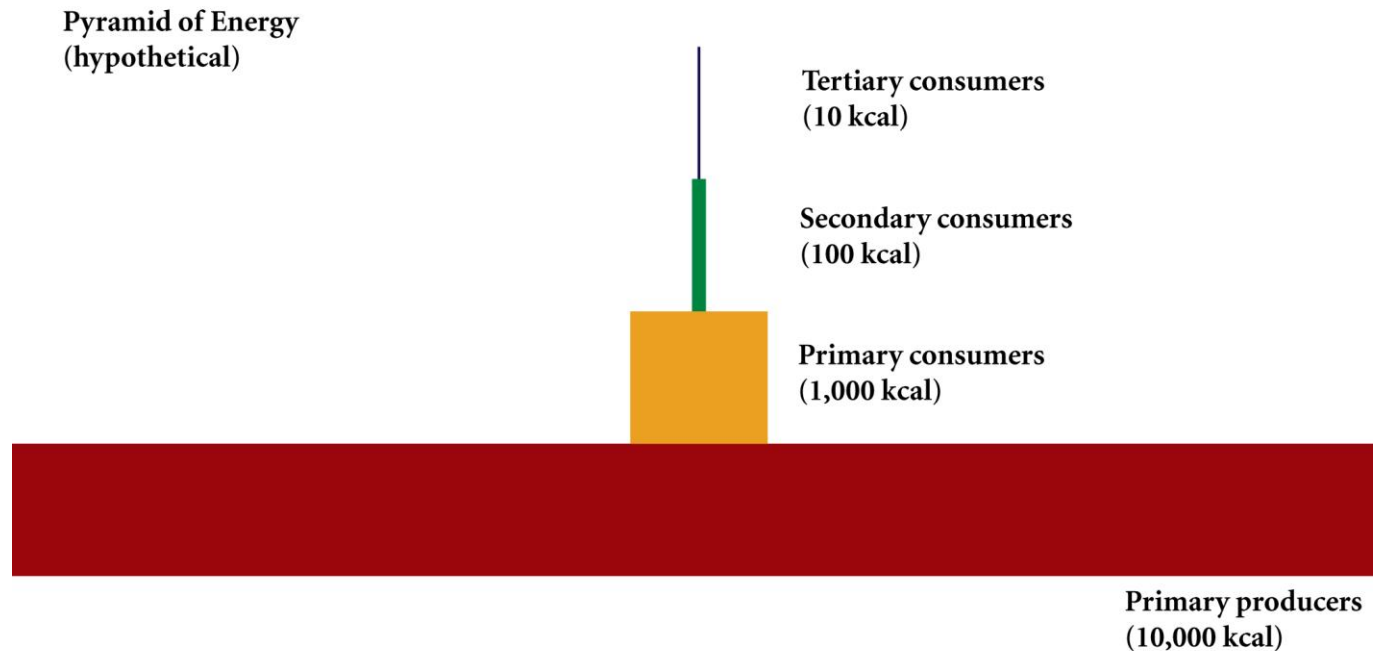


Ecological Pyramids of Biomass



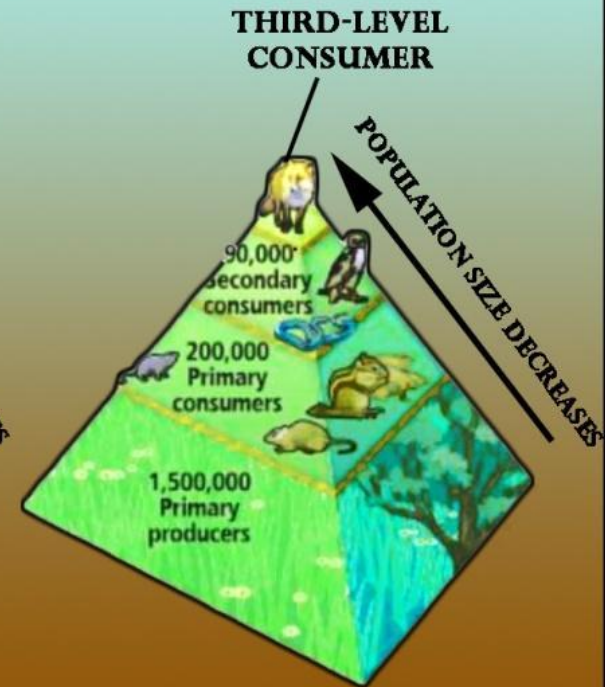
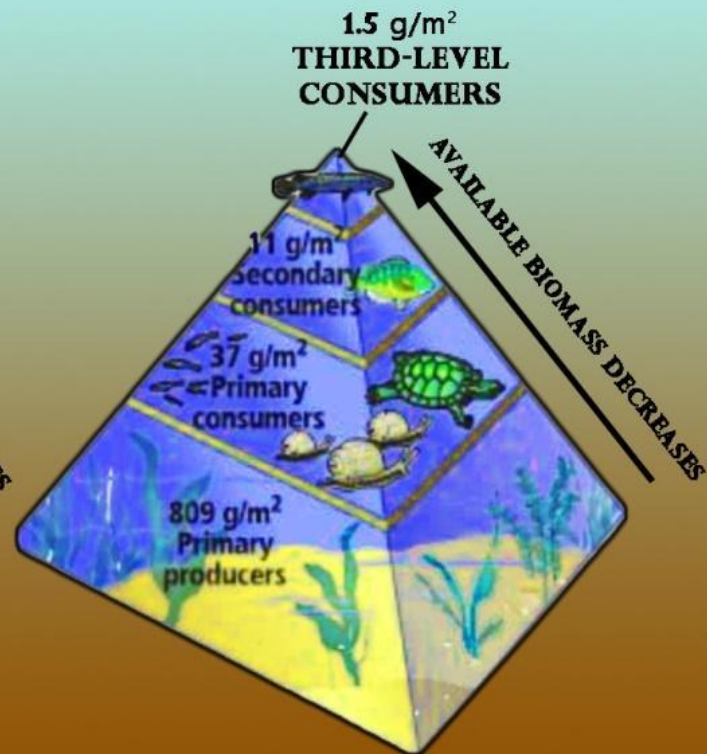
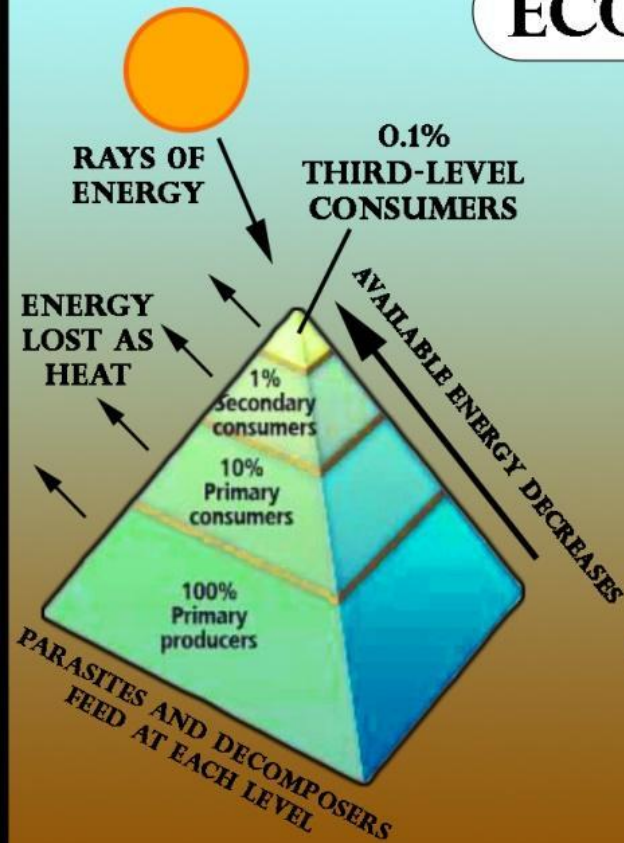
The total dry weight of organisms (standing crop) in a particular trophic level is referenced as biomass. Most pyramids follow the typical pattern of narrowing at each level, however in some aquatic ecosystems, the pyramid may be inverted. In the example, phytoplankton grow and reproduce so rapidly that they can support a large population of zooplankton even though at any one time, the biomass of phytoplankton is smaller than that of the zooplankton.

Ecological Pyramids of Energy



Energy in ecosystems flows from producers (photosynthetic organisms) to consumers (herbivores and carnivores). Ecological pyramids of energy usually depict the amount of living material (or its energetic equivalent) that is present in different trophic levels. In this diagram, energy is depicted in kilocalories. **Primary producers convert only about 1% of the energy in available sunlight. The average amount of energy that is available to the next trophic level is about 10%.** Because so much energy is utilized in building and maintaining organisms, food chains (series of feeding relationships) are usually limited to just three or four steps. Pyramids of energy cannot be inverted.

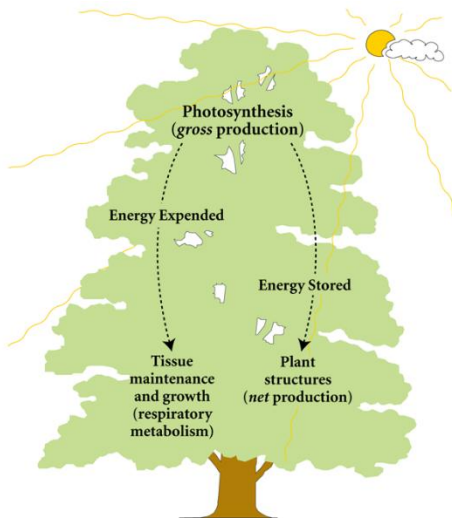
ECOLOGICAL PYRAMIDS



PYRAMID OF ENERGY

PYRAMID OF BIOMASS

PYRAMID OF NUMBER



Primary Productivity

The conversion of light energy to chemical energy is called 'Gross Primary Production (GPP)'. Plants use the energy captured in photosynthesis for maintenance and growth. The energy that is accumulated in plant biomass is called 'Net Primary Production (NPP)'.

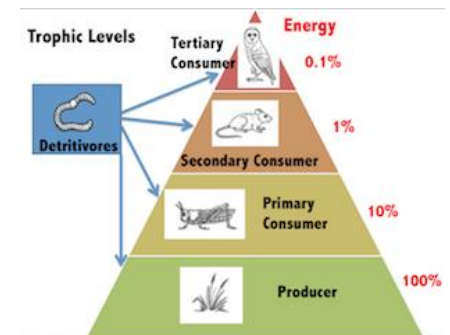
$$\text{NPP} = \text{GPP} - \text{Respiration rate}$$

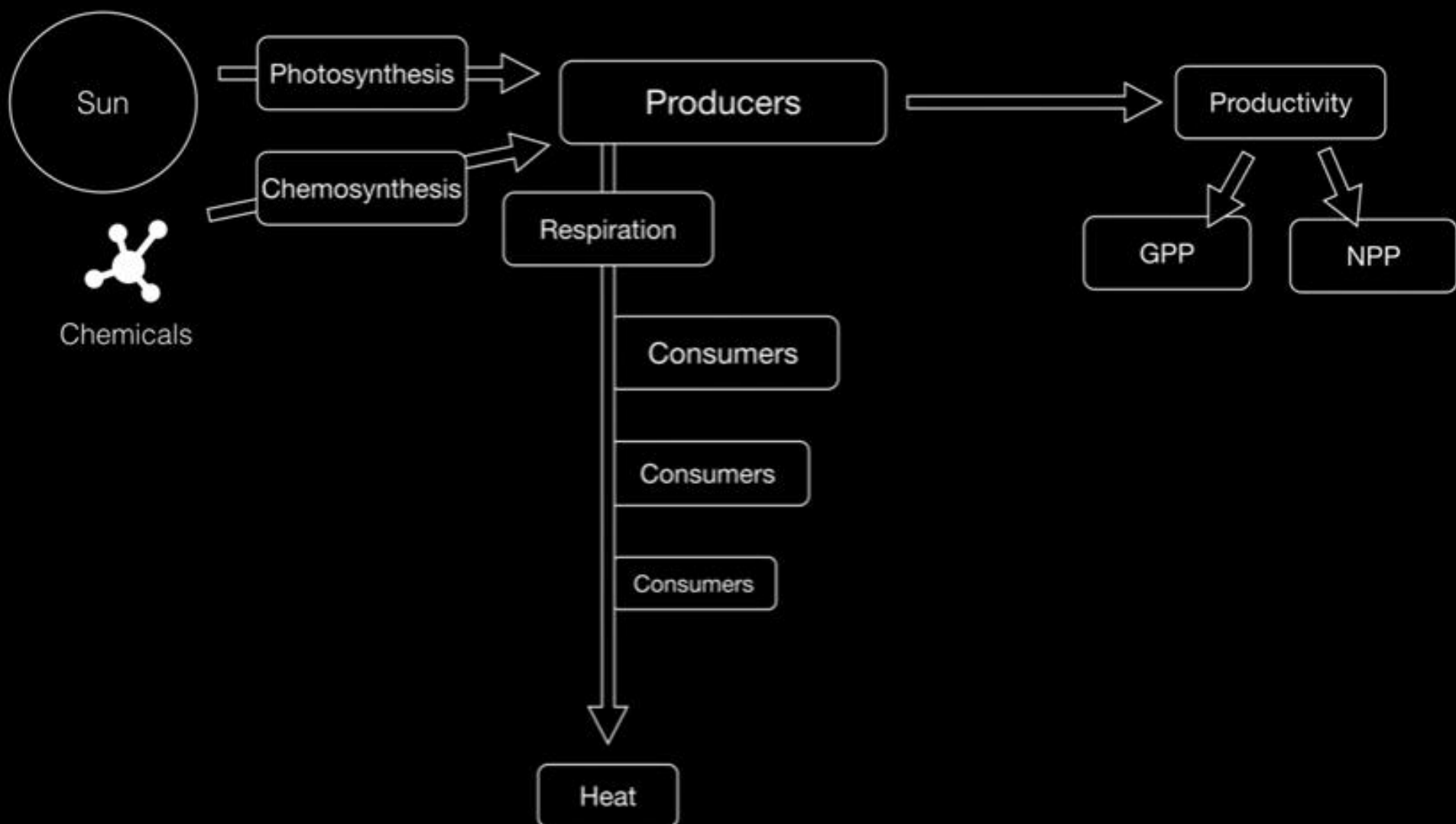
GPP is the rate at which producers convert solar energy into chemical energy as biomass, and NPP is the rate at which energy for use by consumers is stored in new biomass (available to consumers).

The producers must use some of the total biomass they produce for their own respiration.

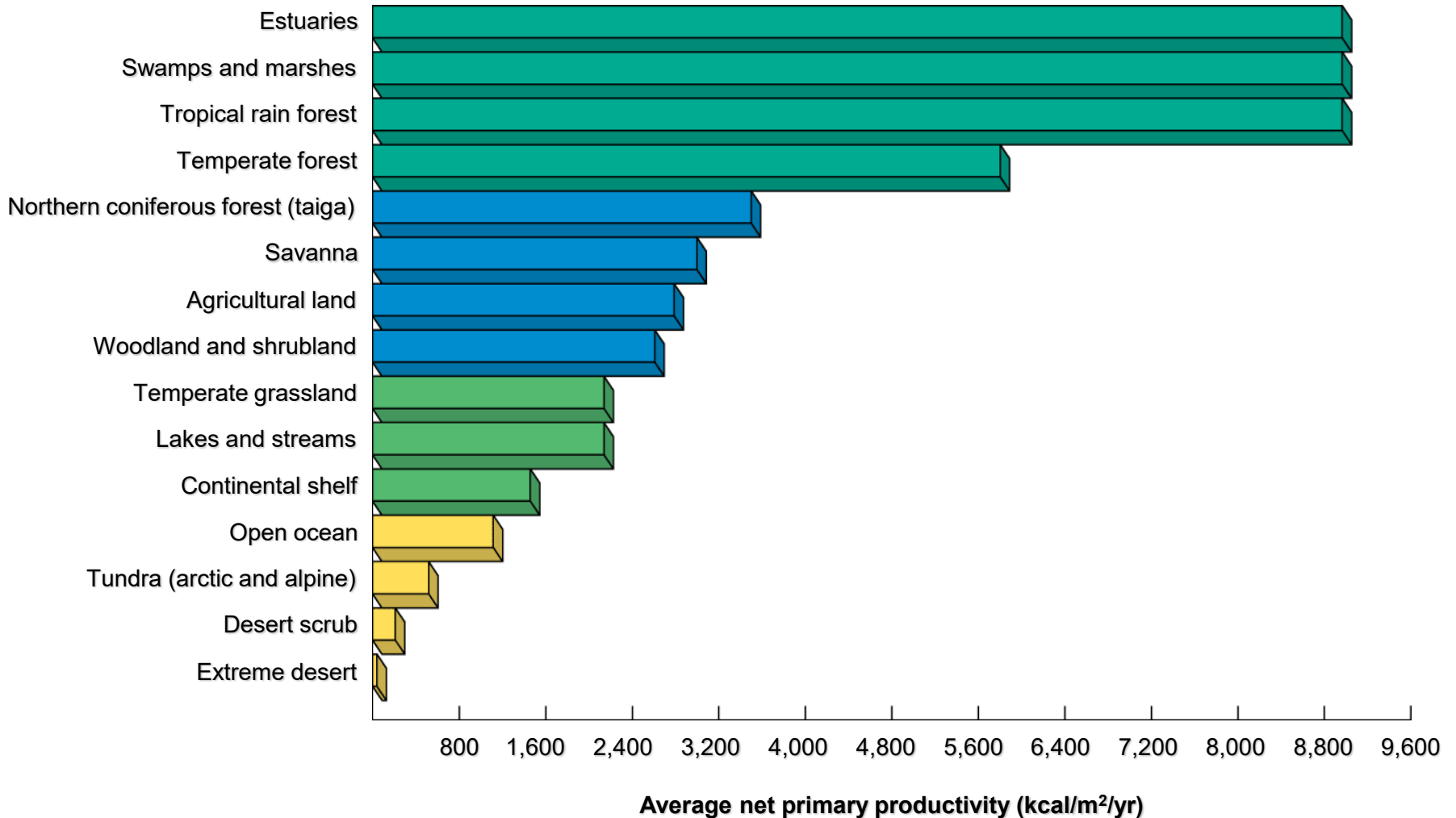
Units:

Kcal/m²/yr or g/m²/yr



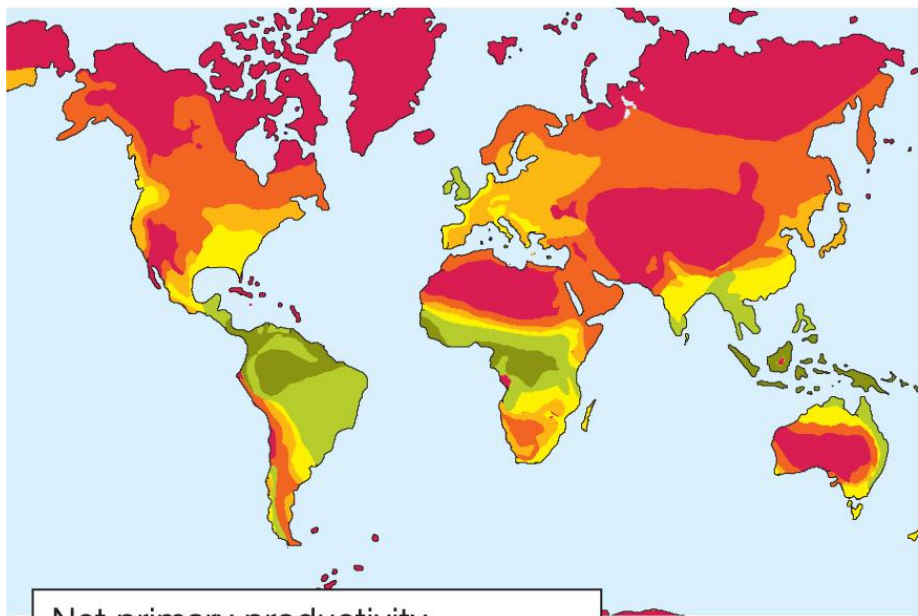


What are the Most Productive Ecosystems?

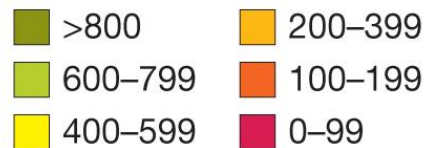


Net Primary Productivity

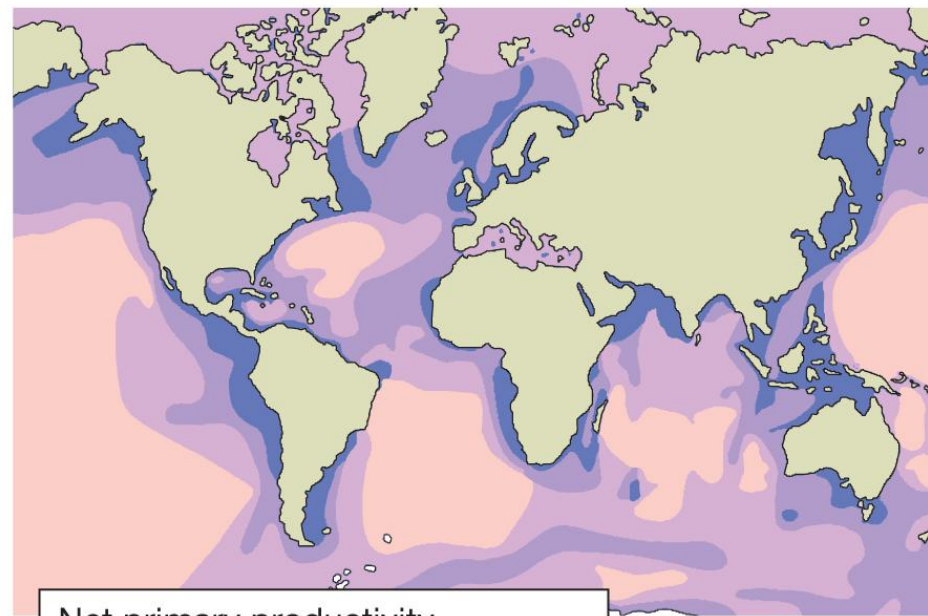
NPP in terrestrial ecosystems



Net primary productivity
(grams per square meter per year)



NPP in aquatic ecosystems



Net primary productivity
(grams per square meter per year)



<https://www.youtube.com/watch?v=0ZOvqYypOuo>

Reading:

FOOD CHAINS AND FOOD WEBS



<https://www.epa.gov/sites/production/files/documents/foodchainsandfoodwebs.pdf>